00-S-550B
December 1, 1978
SUPERSEDING
Fed. Spec. 00-S-550A
June 16, 1969

FEDERAL SPECIFICATION

SOCKET, TAPER-SHANK TOOL (MACHINE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers sockets for holding various types of taper-shank cutting tools in machine spindles and turrets.

1.1.1 Federal specification coverage. Federal specifications do not cover all types and sizes of the commodity indicated by the title of the specifications, but are intended to cover only those most generally used by the Federal Government.

1.2 Classification.

1.2.1 Types, classes, and styles. Sockets shall be of the following types, classes, and styles as specified (see 6.1):

Type I - Chucking.
Type II - Extension.
Type III - Reducing socket (sleeve).
  Class 1 - American Standard taper.
  Class 2 - Brown and Sharpe taper.
  Class 3 - Flatted (Use-Em-Up)
  Class 4 - Adapter.
    Style A - Brown and Sharpe to American Standard taper.
    Style B - American Standard to Brown and Sharpe taper.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

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(Copies of Military specifications and standards required be contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American National Standards Institute, Inc., (ANSI) Standard:

ANSI B5.10 - Machine Tapers

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

Uniform Classification Committee, Agent:

Uniform Freight Classification

(Application for copies should be addressed to Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., Traffic Department, 1616 P Street NW, Washington DC 20036.)

3. REQUIREMENTS

3.1 Material. Sockets shall be of a good grade of recycled, virgin or reclaimed steel of uniform quality and shall conform in composition, heat treatment, and serviceability to the standard practices of reputable manufacturers producing sockets of the types specified herein.

3.2 Design. All sockets shall be furnished with a drift keyslot. Socket shank tapers, hole tapers, drift keyslots, and tangs shall conform to the applicable dimensional requirements of USAS B5.10.

3.3 Concentricity. The shank taper shall be ground concentric with the taper hole. All points on the external taper shall be concentric with the internal taper within a tolerance of 0.002-inch total indicator reading when tested as specified in 4.5.2, except for type II where 0.005-inch total indicator reading will be allowed.

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3.4 Hardness. Sockets shall be furnished hardened or unhardened, as specified (see 6.1). Sockets specified as unhardened shall have the tangs heat treated to a minimum Rockwell "C" hardness of 30 or shall be undercut or relieved at the option of the manufacturer. Sockets specified as hardened shall have the tangs and hole ends heat treated to a minimum Rockwell "C" hardness of 50, except that type III, class 1 with standard taper-hole sizes of No. 4 and larger which shall be fully hardened.

3.5 Finish. The shank taper on all sockets shall be ground. The hole taper shall be reamed, except for fully hardened type III, class 3 sockets which shall be finished commercial standard. Type III, class 1 standard taper holes or No. 4 and larger shall be ground.

3.6 Marking for identification. Each socket shall be permanently marked with the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined. They shall also be plainly and permanently marked with the number of the taper hole, followed by the number of taper shank or the diameter of the shank in the case of type I. When the taper is either a Brown and Sharpe or a flatted (Use-Em-Up) taper the letters "B&S" or "F" shall be included following the number of taper.

3.7 Type I, chucking sockets. Type I sockets shall be furnished with a straight shank, a standard taper hole and a drift keyslot. The straight shank shall be ground and the taper hole shall be ground or reamed. The sockets shall be furnished with a ground hole when a fully hardened condition is specified. They shall conform to the dimensions and tolerances of table I for the nominal size specified (see 6.1) and shall be similar to figure 1.

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Table I. Type I, chucking sockets

Nominal size ANSI B5.10		Diameter of shank ±0.001 per inch A	Length of sockets ±1/16 B	Taper hole standard taper No. C
Taper hole	Shank diameter			
	Inches	Inches	Inches	
1	1	1.000	3 1/2, 4, 4 3/4	1
1	1 1/4	1.250	3 1/2, 4, 4 3/4	1
1	1 1/2	1.500	3 1/2, 4, 4 3/4, 6	1
2	1	1.000	3 1/2, 4, 4 3/4	2
2	1 1/4	1.250	3 1/2, 4, 4 3/4	2
2	1 1/2	1.500	3 1/2, 4, 4 3/4, 6	2
2	1 3/4	1.750	4, 4 3/4, 6	2
2	2	2.000	4, 4 3/4, 6	2
3	1 1/4	1.250	3 1/2, 4, 4 3/4,	3
3	1 1/2	1.500	3 1/2, 4, 4 3/4, 6	3
3	1 3/4	1.750	4, 4 3/4, 6	3
3	2	2.000	4, 4 3/4, 6	3
4	1 1/2	1.500	3 1/2, 4, 4 3/4, 6	4
4	1 3/4	1.750	4, 4 3/4, 6	4
4	2	2.000	4, 4 3/4, 6	4

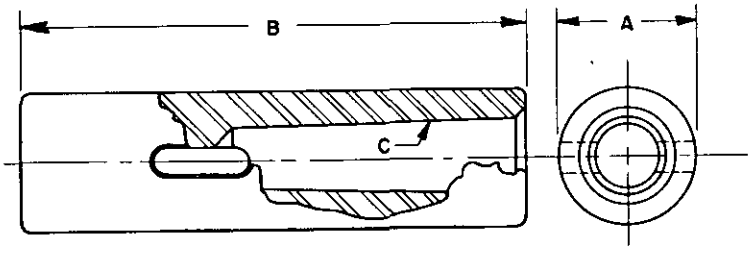


Figure 1. Type I, chucking socket.

3.8 Type II, extension sockets. Type II sockets shall have a standard taper shank, an extension body, a standard taper hole and a drift keyslot. They shall conform to the dimensions and tolerances shown in table II for the nominal size specified (see 6.1), and shall be similar to figure 2.

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Table II. Type II, extension sockets

Nominal size ANSI B5.10		Taper hole standard taper number A	Shank standard taper number B	Overall length (min.) C	Body diameter (min.) D
Hole number	Outside number				
1	1	1	1	5 1/2	7/8
1	1	1	1	9	7/8
1	1	1	1	12	7/8
1	1	1	1	18	7/8
1	2	1	2	6	1
1	2	1	2	7 5/8	1
1	3	1	3	7	1
1	4	1	4	8	1
1	5	1	5	9	1
2	2	2	2	6 3/4	1 1/8
2	2	2	2	9	1 1/8
2	2	2	2	12	1 1/8
2	2	2	2	18	1 1/8
2	3	2	3	7 1/2	1 1/8
2	4	2	4	8 1/2	1 1/8
2	5	2	5	9 3/4	1 1/8
3	2	3	2	7 1/2	1 3/8
3	3	3	3	8	1 3/8
3	3	3	3	9	1 3/8
3	3	3	3	12	1 3/8
3	3	3	3	18	1 3/8
3	4	3	4	9 1/2	1 3/8
3	5	3	5	10 3/4	1 3/8
4	3	4	3	9 1/4	1 13/16
4	4	4	4	10 1/2	1 13/16
4	4	4	4	12	1 13/16
4	4	4	4	18	1 13/16
4	5	4	5	11 3/4	1 13/16
4	6	4	6	14 1/4	1 13/16
5	4	5	4	11 5/8	2 5/16
5	5	5	5	12 7/8	2 5/16
5	6	5	6	15 1/4	2 7/16

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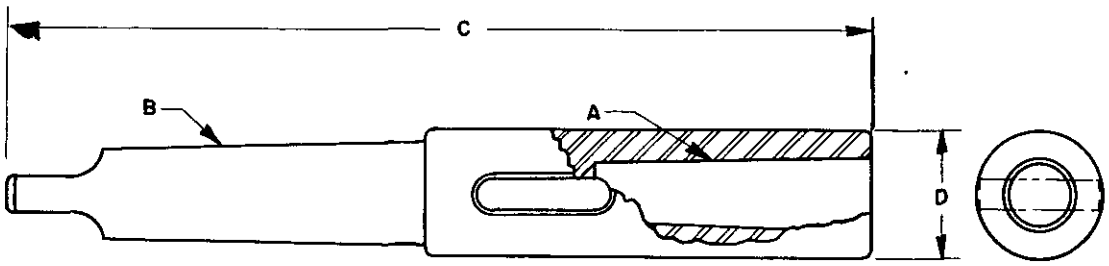


Figure 2. Type II, extension socket.

3.9 Type III, reducing socket (sleeve).

3.9.1 Class 1, American standard taper. Class 1 sockets shall have a standard taper shank, a standard taper hole and a drift keyslot. They shall conform to the dimensions and tolerances shown in table III for the nominal size specified (see 6.1), and shall be similar to figure 3.

3.9.2 Class 2, Brown and Sharpe taper. Class 2 sockets shall have a Brown and Sharpe taper shank, Brown and Sharpe taper hole, and a drift keyslot. Unless otherwise specified in the contract or order, they shall be furnished with a tang drive. If so specified, they shall have a threaded hole for a draw-in bolt of the size specified. They shall conform to the dimensions and tolerances shown in table IV for the nominal size specified (see 6.1) and shall be similar to figure 3.

Table III. Type III, reducing sockets, class 1, and class 3 American Standard taper

Nominal size ANSI B5.10		Taper hole standard taper number A	Shank standard taper number B	Overall length ± 1/8 C
Hole number	Outside Number			
				Inches
1	2	1	2	3 9/16
1	3	1	3	3 15/16
1	4	1	4	4 7/8
1	5	1	5	6 1/8
2	3	2	3	4 7/16
2	4	2	4	4 7/8
2	5	2	5	6 1/8
3	4	3	4	5 3/8
3	5	3	5	6 1/8
4	5	4	5	6 5/8
4	6	4	6	8 5/8
5	6	5	6	8 5/8

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Table IV. Type III, reducing sockets classes 2 and 3 Brown and Sharpe taper

Nominal size		Taper hole Brown/Sharpe taper number A	Shank Brown/Sharpe taper number B	Overall length C
Hole number	Outside number			
				Inches
Brown and Sharpe 4	Brown and Sharpe 7	1/4	7	4 5/8
Brown and Sharpe 5	Brown and Sharpe 7	T/5	7	4 5/8
Brown and Sharpe 5	Brown and Sharpe 9	T/5	9	5
Brown and Sharpe 5	Brown and Sharpe 10	T/5	10	7 1/15
Brown and Sharpe 7	Brown and Sharpe 9	7	9	5
Brown and Sharpe 7	Brown and Sharpe 10	7	10	7 1/16
Brown and Sharpe 7	Brown and Sharpe 11	7	11	7 5/8
Brown and Sharpe 9	Brown and Sharpe 10	9	10	7 1/16
Brown and Sharpe 9	Brown and Sharpe 11	9	11	7 5/8
Brown and Sharpe 9	Brown and Sharpe 12	9	12	8 1/8
Brown and Sharpe 10	Brown and Sharpe 11	10	11	8 5/8
Brown and Sharpe 10	Brown and Sharpe 12	10	12	9

1/ Brown and Sharpe taper holes No. 4 and 5 shall be fitted for a shank without tang.

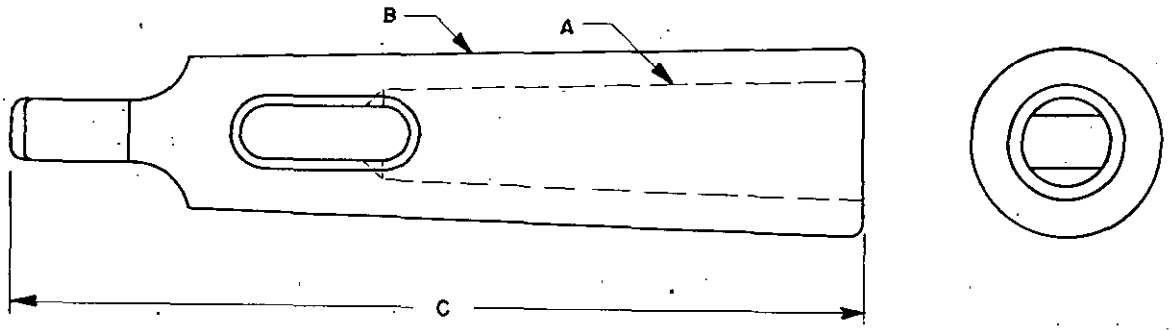


Figure 3. Type III, reducing socket, classes 1, 2, and 4.

3.9.3 Class 3, flatted (Use-Em-Up). Class 3 sockets shall be designed to hold cutting tools with broken shanks and damaged tangs which have been properly flatted on the shank. They shall be similar to type III, class 1 except that the taper hole shall have a flatted side on about 1/4 of the circumference for the full length of the taper. They shall have a long drift keyslot to permit the removal of tools with damaged shanks. They shall conform to the dimensions of tables III and IV for the size specified (see 6.1), and shall be similar to figure 4. The marking of the flatted hole size shall include the letter "F" following the size.

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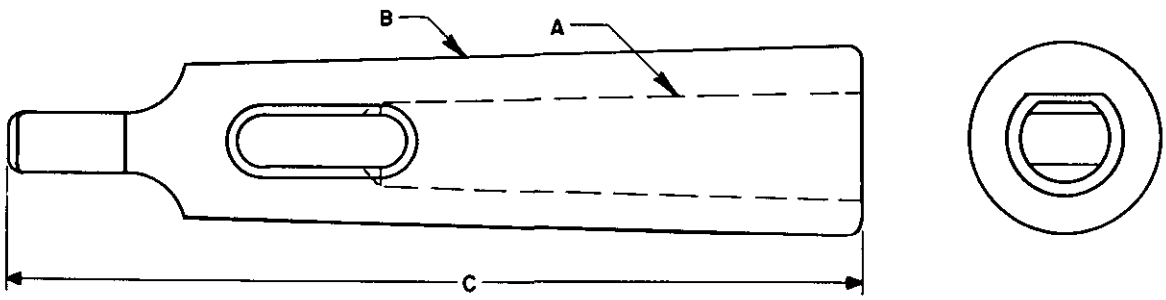


Figure 4. Type III, reducing socket, class 3

3.9.4 Class 4, adapter (figure 3). Class 4 socket adapters shall be designed to use Brown and Sharpe taper-shank tools in machines with standard taper spindles or to use standard taper-shank tools in machines or attachments equipped with Brown and Sharpe spindles. The overall length shall be in accordance with the manufacturer's established practice.

3.9.4.1 Style A. Brown and Sharpe taper holes to standard taper shanks. Style A adapter reducing sockets shall be furnished with a Brown and Sharpe taper hole and a standard taper shank as specified (see 6.1). They shall have a drift keyslot.

3.9.4.2 Style B, standard taper holes to Brown and Sharpe taper shanks. Style B adapter reducing sockets shall be furnished with a standard taper hole and a Brown and Sharpe taper shank, as specified (see 6.1). Unless otherwise specified, the Brown and Sharpe taper shank shall be furnished with a tang drive. If so specified they shall have a threaded hole for draw-in bolt of the size specified. They shall be furnished with a drift keyslot.

3.10 Workmanship. All sockets shall be free from imperfections which may affect their appearance, serviceability, or durability. Plating or coating, when required, shall completely cover all surfaces with no evidence of chipping or flaking. All surfaces and edges shall be free from defects which may be injurious to the user.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Quality conformance inspection. Quality conformance inspection shall be applied to production items being offered for acceptance under the contract. Unless otherwise specified in Section 6, quality conformance inspection shall consist of (a) and (b) as follows. Failure of any unit to pass an examination, test or inspection shall be cause for rejection of the unit.

(a) Examination (4.2.2).
(b) Tests (4.2.3).

4.2.1 Sampling. All sockets of the same type, class and style offered for delivery at one time shall be considered a lot for purposes of quality conformance inspection. A random sample of sockets shall be selected from each lot offered for inspection in accordance with MIL-STD-105 at General Inspection Level and the Acceptable Quality Level (AQL) specified.

4.2.2 Examination. Each socket shall be examined to determine compliance with Section 3 herein and the contract relative to its type, class and style as specified. The inspection shall be at General Inspection Level II and acceptance on an AQL of 1.0. (see 3.1, 3.2, 3.5, and 3.6)

4.2.3 Tests. Each socket shall be tested to determine compliance with Section 3 herein and the contract. Inspection shall be at General Inspection Level S-4 and acceptance on an AQL of 1.5.

4.2.3.1 Concentricity tests. Tests for concentricity shall be made in the following manner: A mandrel accurately ground on centers to fit the inside taper of the socket being tested shall be also ground straight for an additional six inches to approximately the size of the large end of the taper. After the taper end of the mandrel is inserted in the piece being inspected, the straight section of the mandrel shall be supported in a "V" block using a steel ball in the center and a stop to prevent axial movement. The mandrel shall be rotated and no point on the external taper shall show a greater run out on a dial indicator than specified in 3.3.

4.2.3.2 Hardness tests. The hardness of sockets shall be determined using a Rockwell hardness tester ("C" scale). The hardness requirements for sockets are specified in Sections 6.1 and 3.4. The heat treated tangs of unhardened sockets shall have a minimum hardness of C-30 or shall be undercut or relieved. The tangs and holes of sockets which have specified heat treat shall have a minimum hardness of C-50. Type III, Class I sockets with standard taper-hole sizes of No. 4 and larger shall be fully hardened (see 3.4).

4.3 Examination of preparation for delivery. An examination shall be made to determine compliance with the requirements of Section 5. The sample unit shall be one shipping container fully prepared for delivery. Sampling shall be in accordance with MIL-STD-105. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of percent defective.

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5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing. Sockets of one description only shall be preserved and packaged in accordance with normal commercial practice and packed to assure acceptance by common carrier and provide product protection against loss and damage during multiple shipments, handling and storage. The shipping container shall be in compliance with National Motor Freight Classification and Uniform Freight Classification.

5.2 Marking. Marking shall be as specified in the contract or order.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

(a) Title, number, and date of this specification.
(b) Type, class, style and sizes of sockets required (see 1.2 and applicable tables of sizes).
(c) Whether hardened or unhardened (see 3.4).

Military Custodians:
Army - AL
Navy - YD
Air Force - 99

User activity:
Army - AV

Preparing activity:
Army - AL

Civil Agency Coordinating Agency:
GSA-FSS

Project No. 3460-0073

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NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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A. GIVE PARAGRAPH NUMBER AND WORDING.

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